Ultralow noise microwave generation with fiber-based optical frequency comb and application to atomic fountain clock

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We demonstrate the use of a fiber-based femtosecond laser locked onto an ultrastable optical cavity to generate a low-noise microwave reference signal. Comparison with both a cryogenic sapphire oscillator (CSO) and a titanium-sapphire-based optical frequency comb system exhibit a stability of about 3×10^{-15} between 1 and 10 s. The microwave signal from the fiber system is used to perform Ramsey spectroscopy in a state-of-the-art cesium fountain clock. The resulting clock is compared to the CSO and exhibits a stability of $3.5\times10^{-14}\tau^{-1/2}$.
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Atomic fountain primary frequency standards based on cold atoms are the most widely used high accuracy atomic clocks.[1] About ten fountains currently participate in the definition of the SI second at a level of 10^{-15} or better.[2–5] In addition, accurate and stable atomic fountain clocks can also perform high precision fundamental physics tests.[6–8]

State-of-the-art microwave atomic fountain clocks[5] exhibit quantum projection noise (QPN) (Ref. 9) with a limited short-term stability well below 10^{-13} at 1 s integration time. However, the intrinsic phase noise of the microwave signal used as interrogation oscillator for these atomic standards degrades performances from the fundamental QPN limit via the Dick effect.[10,11] Therefore, the realization of extremely low noise microwave oscillators is of prime importance for frequency standards to reach high stability.[9] Other applications of low noise microwave sources include radar, deep space navigation,[12] and ultrahigh resolution very-long-baseline interferometry (VLBI).[13]

The interrogation oscillator for the LNE-SYRTE atomic fountain clocks is currently a liquid helium cryogenic sapphire oscillator (CSO) at 11.932 GHz.[14] A CSO was until recently the only available technology allowing QPN limited stability of fountain clocks at a few 10^{-14} at 1 s. The cost of operation and maintenance associated with cryogenic cooling make it desirable to find an alternative technique. Optical ultrastable reference cavities,[15,16] on the other hand, offer reliable and low maintenance high spectral purity source. Transfer of the stability of such optical reference (typically in the lower 10^{-15} range at 1 s) to the microwave domain by use of a titanium-sapphire-based optical frequency comb (TSOFC) has been demonstrated with a residual instability of 6.5×10^{-16} at 1s.[17] However, for long term and reliable operation, fiber-based optical frequency combs (FOFC) are more desirable. Lipphardt *et al.*[18] recently demonstrated microwave generation at the level of instability of 1.2×10^{-14} at 1 s with such a system.

Here we present a different technique to generate a low noise microwave from a FOFC and demonstrate an instability in the low 10^{-15} range at 1 s by comparisons with both a CSO and a TSOFC generated microwave. The low noise microwave signal at 11.932 GHz from the FOFC was used as a replacement of the CSO signal in our frequency synthesis system[19] to perform Ramsey spectroscopy in a cesium fountain and was locked to the atomic signal. The resulting clock was characterized against the CSO.

Essential to this work was our designing and implementing very low vibration sensitivity optical cavities.[20] The measured frequency instability of the beat-note signal between two cw fiber lasers at 1542 nm stabilized on two independent ultrastable cavities is below 2×10^{-15} at 1 s.[21] A commercial erbium-doped fiber femtosecond laser[22] of repetition rate $f_{\mathrm{rep}}\simeq250$ MHz, with inbuilt f-2f interferometer for measuring the carrier-envelope offset frequency f_0,[23] is locked onto one of these reference lasers. The lock technique is as follows: a 30 mW output port (100 nm spectral bandwidth) from the oscillator is sent through a fibered Bragg grating at 1542 nm whose reflected light (1 nm spectral bandwidth) is directed through a circulator to a fibered 50/50 power combiner where it is mixed with the ultrastable reference light of optical frequency ν_{cw}. The resulting beat-note signal $f_b=\nu_{\mathrm{cw}}-N\times f_{\mathrm{rep}}-f_0$ (with N as a large integer) is detected on a photodiode and mixed with f_0. After filtering, the relevant sideband produces a frequency $\nu_{\mathrm{cw}}-N\times f_{\mathrm{rep}}$ independent of f_0. This signal is cleaned by a tracking oscillator filter (2 MHz bandwidth), divided by 128, and mixed with a reference frequency synthesized by a direct digital synthesizer (referred as DDS1) (see Fig. 1) to produce a phase error signal. This signal acts on the pump-power controller of the femtosecond laser through an optimized phase-lock loop filter. The servo bandwidth is 120 kHz, which, combined with the division factor of 128, allows robust and reliable servo-locking. The high gain of the loop between 1 Hz and 1 kHz allows the noise to be limited, in principle, to that of the reference cw laser. Once locked onto the optical reference ν_{cw}, the repetition rate f_{rep} (and all its harmonics) reproduces its ultrahigh stability, transferred in the microwave domain. Our optical reference laser exhibits a long-term fractional frequency drift of a few 10^{-15} s^{-1}, which is removed by a constant feed-forward linear ramp on DDS1.

To generate microwave signals, the transmitted output of the Bragg grating (9 mW) containing all the spectrum is sent

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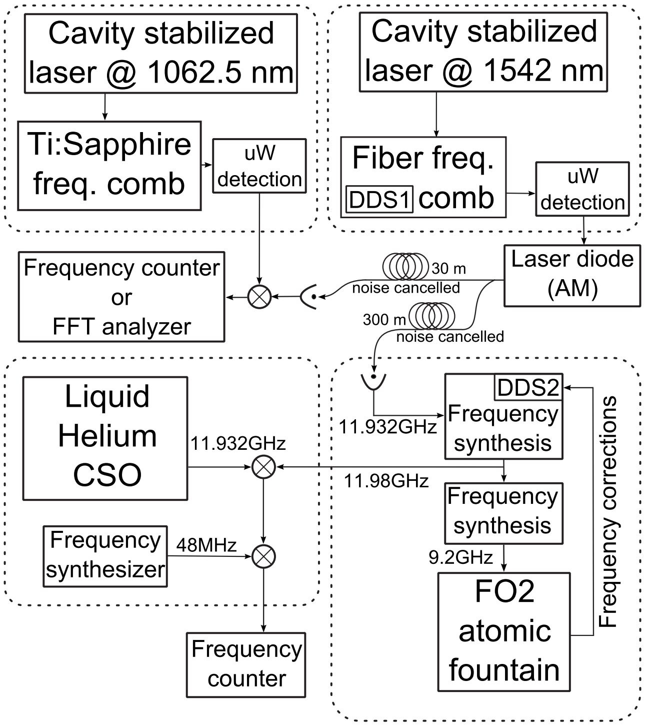

FIG. 1. Schematic of the experimental system. CSO denotes liquid helium-cooled cryogenic sapphire oscillator, DDS denotes direct digital synthesizer, FFT denotes fast Fourier transform analyzer, and uW denotes microwave signal.

to a fast InGaAs pigtailed photodiode (Discovery model DSC40S). The output signal of the detector (7.6 mA in total photocurrent) contains all the harmonics of the repetition rate, up to 20 GHz. In order to characterize and use the microwave from this FOFC in distant laboratories, the harmonic of interest (−27 dBm per peak in the 9–12 GHz range) is amplified and transmitted to distant laboratories by optical fiber link using a round trip noise-compensation scheme similar to Ref. 24.

In a first experiment, the harmonic of the FOFC's repetition rate near 9.2 GHz was sent to a nearby laboratory about 30 m away. There, it was compared to a 9.2 GHz microwave signal generated by a TSOFC (repetition rate $\simeq$770 MHz) locked onto a separate ultrastable laser operating at 1062.5 nm.[20] The TSOFC uses a similar locking technique as the FOFC, although with a higher bandwidth of about 400 kHz. Figure 2 shows a phase noise measurement of the FOFC-TSOFC beat-note signal and Fig. 3 its Allan deviation (400 Hz measurement bandwidth).

In a second experiment, the FOFC's repetition rate harmonic near 11.932 GHz was sent to the CSO/FO2 fountain laboratory 300 m away, and there compared to the 11.932 GHz CSO signal. The Allan deviation of the beat-note signal is shown in Fig. 3 (10 Hz measurement bandwidth). Both CSO-FOFC and TSOFC-FOFC comparisons give a phase noise power spectral density of approximately $10^{-9}/f^2[\mathrm{rad}^2/\mathrm{Hz}]$ (at 9.2 GHz) for Fourier frequencies f in the 0.1 –10 Hz range, and an Allan deviation of about $(3-4)\times 10^{-15}$ at 1 s integration time (see Figs. 2 and 3). These performances, among the very best for microwave sources, along with the reliability and robustness of the fiber-based system, qualifies it as an excellent microwave source for long-term operation of atomic fountain clocks.

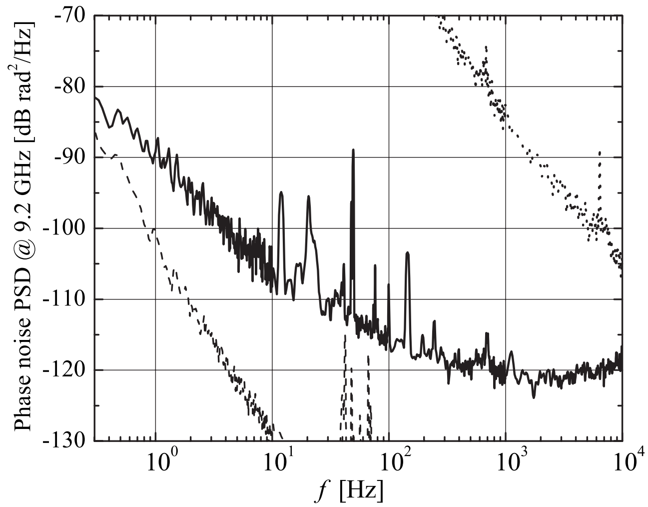

FIG. 2. Phase noise power spectral density at 9.2 GHz of the beatnote between the microwaves generated by the FOFC and the TSOFC. Dotted line is obtained for a free running FOFC. Dashed line: two ultrastable lasers locked onto 1542 nm independent reference cavities (scaled to 9.2 GHz). Degradation from optical to microwave is attributed mainly to amplitude-to-phase noise conversion (Ref. 17) below $\simeq$1 kHz and to limited servo gain above $\simeq$1 kHz.

In a third experiment, the low phase noise signal at 11.932 GHz generated by our FOFC was used to replace the CSO microwave as interrogation oscillator for the FO2 atomic fountain, as shown in Fig. 1. From 11.932 GHz, our frequency synthesis[19] generates a tunable microwave signal shifted to 11.980 GHz via a computer controlled DDS (referred as DDS2). This signal is further used to generate a low phase noise 9.2 GHz microwave signal for Ramsey spectroscopy of the cold cesium atoms. The sequential operation of the fountain produces frequency corrections every 1.5 s, which are applied to DDS2. The 11.980 GHz signal is thereby locked with a bandwidth of $\simeq$0.2 Hz onto the atomic frequency reference. The resulting primary standard referenced signal at 11.980 GHz was compared to the CSO signal at 11.932 GHz. The 48 MHz difference was bridged by a low noise synthesizer. The comparison yields a fountain's instability of $3.5\times 10^{-14}\tau^{-1/2}$ for integration time τ be-

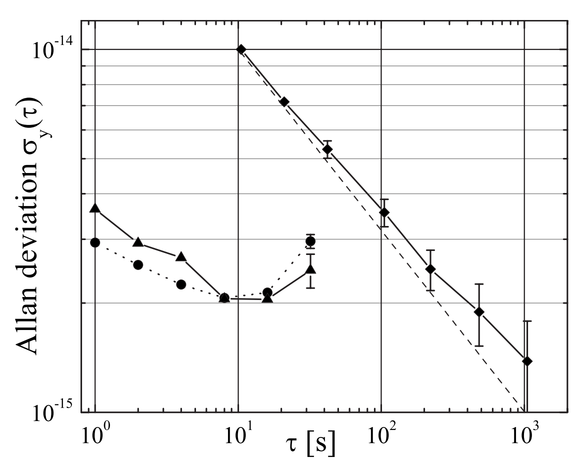

FIG. 3. Circles: Fractional frequency instability (Allan standard deviation) vs integration time of the microwave signal generated by the FOFC against the CSO at 11.932 GHz. Triangles: FOFC vs TSOFC. Diamonds: FOFC locked onto the FO2 atomic signal, compared against the CSO (quadratic drift removed). The latter instability scales as $3.5\times 10^{-14}\tau^{-1/2}$ (dashed line).

tween 10 and 100 s (see Fig. 3; 10 Hz measurement bandwidth). For integration time longer than 100 s, the instability is limited by the flicker floor of $(1-2)\times10^{-15}$ of the CSO. The short-term instability is identical to the one obtained when using the CSO as local oscillator for the FO2 fountain under the same operating conditions. The fountain instability is limited by QPN with the number of atoms of $\simeq 1\times10^{6}$ per shot (1.5 s cycle time). The Dick effect calculated[11] based on the phase noise shown in Fig. 2 is below $5\times10^{-15}\tau^{-1/2}$ for the FO2 fountain current operational parameters.

Continuously operated fiber femtosecond optical frequency comb stabilized onto an ultrastable laser will replace CSO as a flywheel in the near future, removing the use of cryogenics and providing an ultrastable reference in both optical and microwave domains. Furthermore, cross comparisons between FOFC, TSOFC, and CSO microwave generation will allow full characterization and optimization of the three systems and will pave the way to extreme low-noise microwave systems for applications in radar, deep space navigation, and VLBI. Upon completion of this letter, we have become aware of comparable results from Weyers *et al.*[25] with a fountain instability of 7.4×10^{-14} at 1 s.

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